DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026 : PERIODIC ASSESSMENT - 1

SET-1

SUBJECT : SCIENCE (086)

Time: $1\frac{1}{2}$ hrs. MM: 40

Date : _____ Name : _____ Class/Sec X__

General Instructions:

- **Read the following instructions carefully and follow them:**
- **This question paper consists of 18 questions in section A, Section B, Section C, Section D**
- **All questions are compulsory.**
- **Section A consists of 8 questions of 1 mark each.**
- **Section B consists of 4 questions of 2 mark each.**
- **Section C consists of 3 questions of 3 mark each.**
- **Section D consists of 3 questions of 5 mark each.**

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Section – A

Q1. Sanjana is suffering from frequent stomach pain and vomiting. She went to the doctor. The doctor asked her to go for an ultrasound. In the report, a stone was found in the gall bladder. Doctor asked her to remove the gall bladder by operation. But she was reluctant to go for the operation.
What role played by gall bladder in human body?
a) To store bile b) To secrete bile c) To emulsify fat d) To digest fat

Q2. Many processes happen in the bodies of living organisms.
Those processes which involve the building up of complex molecules from simpler ones are called anabolism. Those which involve the breakdown of complex molecules into simpler ones are called catabolism.
Which of the following life processes can be considered as an example of anabolism?
(a) digestion (b) respiration (c) transpiration (d) photosynthesis

Q3. Identify 'x', 'y' and 'z' in the following reaction :

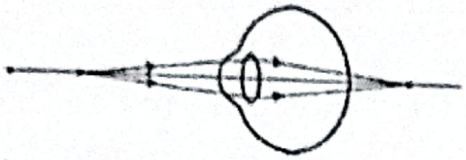
$$2KClO_{3(x)} \xrightarrow{y} 2KCl_{(x)} + O_{2(z)}$$

(a) x = gas; y = reaction condition; z = gas
(b) x = solid; y = liquid; z = gas
(c) x = number of moles of $KClO_3$; y = reaction condition; z = number of molecules of oxygen
(d) x = physical state of $KClO_3$ and KCl ; y = reaction condition, z = physical state of O_2.

Q4. What is the colour of the ash formed when a magnesium ribbon is burnt in the air?
(a) White (b) Black (c) Yellow (d) Pink

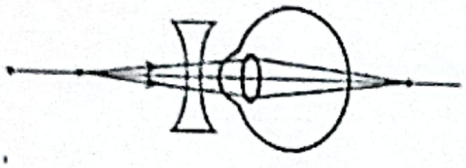
Q5. A student conducts an activity using a flask of height 15 cm and a concave mirror. He finds that the image formed is 45 cm in height. What is the magnification of the image?
(a) -3 times (b) -1/ 3 times (c) 1/ 3 times (d) 3 times

Q6. The image shows the ray diagram of a defective eye.

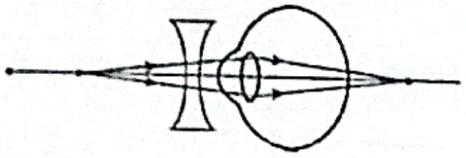


Which option shows the correction of the defect of the eye?

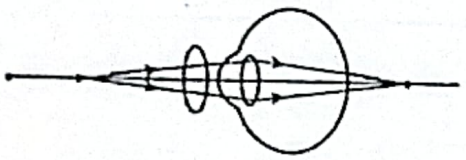
(a)



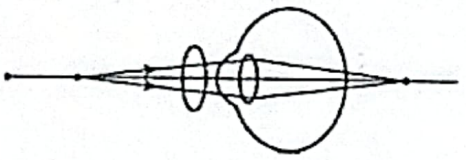
(b)



(c)



(d)



Following questions consist of two statements – Assertion (A) and Reason (R). Answer these questions selecting the appropriate option given below:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Q7. Assertion (A) : **Following is a balanced chemical equation for the action of steam on iron:**
$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
Reason (R): **The law of conservation of mass holds good for a chemical equation.**

Q8. Assertion (A) : **A ray passing through the centre of curvature of a concave mirror after reflection,**

is reflected back along the same path.
Reason (R) : The incident rays fall on the mirror along the normal to the reflecting surface.

Section – B

Q9. Explain about the blood pressure in human beings. Name the instrument used to measure the blood pressure. **2**

Q10. Draw a diagram of human respiratory system and label the following parts. 0.5*4=2
(a) Part where air is filtered by fine hairs and mucus nasal cavity
(b) Part which terminates in balloon-like structure Bronchiuls
(c) Balloon-like structures where exchange of gases takes place. Alveoli
(d) Part which separates chest cavity from abdominal cavity. epiglotis

Q11 A solution of potassium chloride when mixed with silver nitrate solution, an insoluble white substance is formed. Write the chemical reaction involved and also mention the type of the chemical reaction.

Q12. The linear magnification produced by a spherical mirror is +3. Analyse this value and state the (i) type of mirror and (ii) position of the object with respect to the pole of the mirror. Draw a ray diagram to show the formation of image in this case.

Section – C

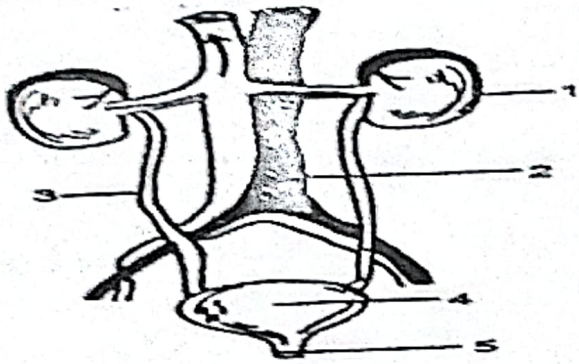
Q13. Lack of oxygen in muscles often leads to cramps among cricketers. Explain. Why?

Q14. Write balanced chemical equations for the following chemical reactions:
(a) Hydrogen + Chlorine → Hydrogen chloride
(b) Lead + Copper chloride → Lead chloride + Copper
(c) Zinc oxide + Carbon → Zinc + Carbon monoxide.

Q15. A student has focused the image of a candle flame on a white screen using a concave mirror. The situation is a given below:
Length of the flame = 1.5 cm
Focal length of the mirror = 12 cm
Distance of flame from the mirror = 18 cm
If the flame is perpendicular to the principal axis of the mirror, then calculate the following:
(a) Distance of the image from the mirror
(b) Length of the image
If the distance between the mirror and the flame is reduced to 10 cm, then what would be observed on the screen? Draw ray diagram to justify your answer from this situation.

Section – D

Q16. Given below is the figure of certain organs and associated parts in the human body, study the same and then answer the questions that follow: **5**



(i) Name the organ systems shown completely or even partially.
(ii) Name the parts numbered 3 and 4.
(iii) Name the structural and functional unit of the part marked '1'.
(iv) Name the two main organic constituents of the fluid that flows down the part labeled '3'.
(v) Name the two major steps involved in the formation of the fluid that passes down the part labeled '3'.

Q17. (A) Read the passage carefully and answer the following questions from (i) to (iv).
The substance which we use for white-washing our houses is calcium oxide. We put calcium oxide in a drum and add water to it slowly. Calcium oxide reacts with water vigorously to form a white solid called slaked lime with the evolution of heat. More water is then added to get slaked lime solution. This slaked lime solution is then applied to the walls of the house with the brush. Since, this process gives a white, shiny appearance to the walls of a house, it is called white-washing and gives a shiny finish to the walls.

(i) What is the chemical formula of slaked lime? (1/2)
(ii) Name the chemical reaction resulting to form slaked lime. (1/2)
(iii) Give the chemical equation for the formation of slaked lime. (1)
(iv) What is the reason behind shining appearance of walls after white washing? (1)

(b) Account for the following:
(i) White silver chloride turns grey in sunlight.
(ii) Corrosion of some metals is an advantage.

Q18. (a) The refractive indices of glass and water with respect to air are 3/2 and 4/3 respectively. If speed of light in glass is 2 × 108 m/s, find the speed of light in water.
(b) A real image 2/3rd of the size of an object is formed by a convex lens when the object is at a distance of 12 cm from it. Find the focal length of the lens.